

Constructing a Leslie Matrix from Fisheries Data

In this assignment, you will use age-structured fisheries data to estimate the fecundity and annual survival probability of a Leslie matrix for female winter flounder.

Learning Objectives

After completing this assignment, you should be able to

- Import and inspect data from an Excel workbook;
- compute catch per unit effort (CPUE);
- estimate age-specific annual survival probabilities S_i ;
- estimate age-specific fecundities F_i ;
- construct a Leslie matrix L from estimated survival probabilities and fecundities;
- evaluate the assumptions and limitations of the model.

Background

The female population is divided into six age classes:

age-1, age-2, age-3, age-4, age-5, age-6+.

Let

$$\mathbf{x}(t) = [x_1(t), x_2(t), x_3(t), x_4(t), x_5(t), x_{6+}(t)]^T$$

denote the age-structured population vector in year t . Here $x_i(t)$ denotes the Catch Per Unit Effort (CPUE), an index of the relative abundance of females in age class i during year t , defined as

$$x_i(t) = \frac{\text{female count in age class } i \text{ during year } t}{\text{number of sampling tows during year } t}.$$

A one-year Leslie matrix model has the form

$$\mathbf{x}(t+1) = L\mathbf{x}(t),$$

where

$$L = \begin{bmatrix} F_1 & F_2 & F_3 & F_4 & F_5 & F_{6+} \\ S_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & S_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & S_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & S_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & S_5 & S_{6+} \end{bmatrix}.$$

The first row contains age-specific fecundities. The subdiagonal entries contain annual survival probabilities between successive age classes.

Data

The workbook `data.xlsx` contains the worksheets

Worksheet	Description
AllFemales	Annual female counts by age class
MatureFemales	Annual mature-female counts by age class
TowCount	Number of sampling tows by year
Temperature	Mean water temperature by year

1 Import and Inspect the Data

Use the starter code below.

```
import pandas as pd
import numpy as np

# -----
# Parameters
# -----
excel_file = "data.xlsx" # input file

# -----
# Read data from Excel
# -----
allfem_counts = pd.read_excel(excel_file, sheet_name="AllFemales",
                              index_col=0)
mature_counts = pd.read_excel(excel_file, sheet_name="MatureFemales",
                              index_col=0)
tow_counts = pd.read_excel(excel_file, sheet_name="TowCount", index_col=0)

allfem_counts.index = allfem_counts.index.astype(int)
mature_counts.index = mature_counts.index.astype(int)
tow_counts.index = tow_counts.index.astype(int)
```

Tasks

- Display the first five rows of each worksheet.
- Record the number of years and age classes.
- Check for missing values.

2 Compute CPUE

For each age-class and each year compute $\text{CPUE}_{i,t} = x_i(t)$. Complete the code below.

```
# TODO:
# Divide each age-class count by the number of tows
# in the corresponding year.

allfem_df =

mature_df =
```

Verify that the CPUE data have the same age-class columns as the original count data.

3 Estimate age-specific survival probabilities

Modeling assumptions

In this project, apparent survival is estimated under the assumptions that

- changes in CPUE primarily reflect mortality, although permanent emigration may also contribute;
- survival is approximately constant over the study period;
- a centered five-year rolling average reduces short-term variability while preserving broader trends.

Use

$$\widetilde{\text{CPUE}}_{i,t}$$

to denote the smoothed CPUE for age class i in year t .

The annual apparent-survival ratio is

$$r_{i,t} = \frac{\widetilde{\text{CPUE}}_{i+1,t+1}}{\widetilde{\text{CPUE}}_{i,t} + \varepsilon},$$

where $\varepsilon > 0$ is a small constant used to avoid division by zero.

The survival estimate is

$$S_i = \min \left(1, \exp \left[\frac{1}{n} \sum_{t=1}^n \log(r_{i,t}) \right] \right),$$

where n is the number of years with valid data.

Starter code

```
smooth_window = 5
eps = 1e-6

# Compute a centered rolling mean.
smoothed_df = allfem_df.rolling(
    window=smooth_window,
    center=True,
    min_periods=1
```

```

).mean()

ages = [1, 2, 3, 4, 5]

# Keep only years for which year t+1 is also available.
valid_t_years = [
    year for year in allfem_df.index
    if (year + 1) in allfem_df.index
]

r_dict = {i: [] for i in ages}

for t in valid_t_years:
    t1 = t + 1

    for i in ages:
        cpue_i_t = smoothed_df.at[t, f"Age{i}"]
        cpue_ip1_t1 = smoothed_df.at[t1, f"Age{i+1}"]

        # TODO:
        # Compute r_{i,t} and append it to r_dict[i].

```

Now aggregate the yearly ratios using a geometric mean.

```

S = {}

for i in ages:
    vals = np.array(r_dict[i])

    # Retain only positive ratios.
    pos = vals[vals > 0]

    if len(pos) > 0:
        # TODO:
        # Compute the geometric mean of the positive ratios.
        gm =

        # Restrict the estimate to the interval [0,1].
        S_i =
    else:
        S_i = 0.0

    S[f"S_{i}"] = round(S_i, 4)

# Terminal age class
S["S_6"] = 0.0

```

Questions

1. Why is a geometric mean used rather than an arithmetic mean?
2. Why is each survival estimate truncated at 1?
3. Why is S_{6+} set to zero in this model?

- Why should we interpret S_i as apparent-survival indices rather than direct estimates of true survival?

4 Estimate Age-Specific Fecundities

Let $M_i(t)$ be the mature-female CPUE in age class i during year t , and let

$$M_{\text{total}}(t) = \sum_{i=2}^{6+} M_i(t).$$

Let $R(t+1)$ denote the number of age-1 fish per tow in year $t+1$. We assume the contribution of age class i to recruitment is proportional to its fraction of the total mature female CPUE in year t .

The age-specific fecundity coefficient is estimated as the geometric mean across all the valid years

$$F_i = \exp \left[\frac{1}{N} \sum_t \log \left(R(t+1) \frac{M_i(t)}{M_{\text{total}}(t)} \right) \right],$$

where N is the number of years used in averaging.

Assume that age-1 females are non-reproductive, so $F_1 = 0$.

Recruitment alignment

Age-1 CPUE in year $t+1$ must be aligned with mature-female CPUE in year t .

```
age_cols = [
    "Age1", "Age2", "Age3",
    "Age4", "Age5", "Age6+"
]

recruit_df = allfem_df[["Age1"]].rename(
    columns={"Age1": "Recruitment"}
).copy()

# Shift the recruitment year backward so Age-1 CPUE
# in year t+1 is paired with mature females in year t.
recruit_df["Year"] = recruit_df.index - 1
recruit_df.set_index("Year", inplace=True)

# Merge by year.
df = mature_df.merge(
    recruit_df,
    left_index=True,
    right_index=True,
    how="inner"
)
```

Starter code

```

# TODO:
# Compute each mature age class as a proportion
# of total mature-female CPUE.

df_prop =

# TODO:
# Multiply the mature proportions by next-year recruitment.

df_scaled =

fecundity_final = {}

for age in age_cols:
    vals = df_scaled[age].values
    vals = vals[vals > 0]

    if len(vals) > 0:
        # TODO:
        # Compute the geometric mean.
        gm =
    else:
        gm = 0.0

    fecundity_final[age] = round(gm, 3)

# Biological rule
fecundity_final["Age1"] = 0.0

```

Questions

1. Which assumptions are required to interpret these quantities as fecundity coefficients?
2. Which age class has the largest estimated fecundity?

5 Construct the Leslie Matrix

Use the estimated fecundities and survival probabilities to assemble the matrix.

```

F = [
    fecundity_final["Age1"],
    fecundity_final["Age2"],
    fecundity_final["Age3"],
    fecundity_final["Age4"],
    fecundity_final["Age5"],
    fecundity_final["Age6+"]
]

L = np.zeros((6, 6))

# TODO:
# Place fecundities in the first row.

```

```
# TODO:
# Place S_1,...,S_5 on the subdiagonal.

# Terminal-age entry
L[5, 5] = S["S_6"]

print(np.round(L, 3))
```

Checks

Verify that L is 6×6 , all fecundities are nonnegative, all survival probabilities lie in $[0, 1]$, the first row contains fecundities, the subdiagonal contains survival estimates.

Interpretation

Write a short report addressing the following questions:

1. Which survival probabilities and fecundities are the largest? What biological processes could explain these patterns?
2. Which assumptions are most restrictive?
3. How might immigration, emigration, or sampling variability affect the matrix entries?
4. Why should the resulting Leslie matrix be interpreted as a simplified model rather than an exact description of the population?

Deliverables

1. The completed Jupyter notebook;
2. The estimated survival probabilities and fecundities;
3. The completed Leslie matrix;
4. A one-page report with the interpretation of the results.

Possible extension

Repeat the annual survival analysis using a centered three-year rolling mean. Compare the resulting survival estimates with those obtained from the five-year rolling mean.